CP Violation from A-Factor Twist Angles of Hyperbolic Holonomy

Marvin L. Gentry[1]

[1] *Independent Researcher*[*]

(Dated: March 14, 2026)

We extend the Iwasawa decomposition program for Standard Model flavor mixing [? ?] to include CP-violating phases. In the companion papers, the CKM and PMNS mixing matrices were derived from the K- and N-factors of the Iwasawa decomposition $\mathrm{PSL}(2,\mathbb{C}) = KAN$ applied to the holonomy of compact hyperbolic 3-manifolds. Here we assign the remaining A-factor: the twist angles $\phi(\gamma) = \mathrm{Im}(\log \lambda(\gamma))$ of loxodromic holonomy elements are the geometric source of CP violation. For the optimal PMNS manifold $m003$ (`OrientableClosedCensus`[1], $H_1 = \mathbb{Z}/5$), the twist angles of the optimal word triple `aa`, `ab`, `aB` predict the CP phase $\delta_{\mathrm{CP}} = \phi_{aa} - \phi_{ab} + \phi_{aB} = 203.5$, within 3.3% of the PDG 2024 value of 197 with no free parameters. We prove that the real Borel construction produces $J = 0$ identically, identify the obstruction as a row-phase invariance theorem, and demonstrate that the full complex construction using imaginary Pauli components of $\log \rho(\gamma)$ generates $J \neq 0$. A census scan identifies the optimal complex word triple and establishes that simultaneous achievement of $\mathcal{F} = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words, providing a concrete prediction for the next scan. The 2:1 translation length ratio $\ell(aa) = 2\ell(aB)$ is proved as a theorem from the group law, with the non-trivial corollary that words a and aB are isospectral on $m003$.

PACS numbers: 12.15.Ff, 14.60.Pq, 02.40.Pc, 11.30.Hv

I. INTRODUCTION

The Iwasawa decomposition $G = KAN$ of $\mathrm{PSL}(2,\mathbb{C})$ provides a natural tripartite structure for Standard Model flavor physics. In the companion papers [? ?], we showed:

- The K-factor (maximal compact, $K \cong \mathrm{SU}(2)$) yields the CKM quark mixing matrix from the holonomy of $m006$ ($H_1 = \mathbb{Z}/5$, fitness $\mathcal{F} = 0.01729$).
- The N-factor (upper unipotent Borel subgroup) yields the PMNS lepton mixing matrix from the holonomy of $m003$ ($H_1 = \mathbb{Z}/5$, fitness $\mathcal{F} = 0.01897$).

The A-factor (diagonal Cartan subgroup) remained unassigned. This paper assigns it: the A-factor encodes CP-violating phases via the twist angles $\phi(\gamma)$ of loxodromic holonomy elements.

A loxodromic element $\gamma \in \pi_1(M)$ has eigenvalue $\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}$, where $t(\gamma) = \ell(\gamma)/2$ is half the translation length and $\phi(\gamma)$ is the twist angle. The A-factor in the Iwasawa decomposition of $\rho(\gamma)$ encodes precisely the modulus $e^{t(\gamma)}$, while the full complex eigenvalue $\lambda(\gamma)$ carries both t and ϕ.

The main results are:

1. **CP phase prediction** (Section **??**): The combination $\phi_{aa} - \phi_{ab} + \phi_{aB} = 203.5$ predicts δ_{CP} within 3.3% of the PDG value, with no free parameters.
2. **Row-phase obstruction theorem** (Section **??**): The real Borel construction has $J = 0$ identically. Row-phase multiplication of L cannot generate $J \neq 0$ because the Jarlskog invariant is invariant under row rephasing.
3. **Complex construction** (Section **??**): Using imaginary Pauli components of $\log \rho(\gamma)$ as complex axis contributions generates $J \neq 0$. Best result: fitness $= 0.049$, $J = -0.017$ at word ordering `Ab/aB/aa`, $\alpha = 0.897$.
4. **Translation length theorem** (Section **??**): $\ell(aa) = 2\ell(aB)$ exactly, with the corollary that words a and aB are isospectral on $m003$.
5. **Prediction** (Section **??**): Simultaneous $\mathcal{F} = 0.019$ and $J = J_{\mathrm{PDG}}$ requires length-3 holonomy words.

II. THE A-FACTOR AND LOXODROMIC TWIST ANGLES

A. Iwasawa decomposition review

Every element $g \in \mathrm{PSL}(2,\mathbb{C})$ decomposes uniquely as $g = k \cdot a \cdot n$, where $k \in K = \mathrm{SU}(2)/\{\pm I\}$, $a \in A$, $n \in N$. The A-factor is

$$a = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \quad t \in \mathbb{R}. \tag{1}$$

For a loxodromic element γ with holonomy $\rho(\gamma) \in \mathrm{PSL}(2,\mathbb{C})$, the eigenvalues of $\rho(\gamma)$ are λ, λ^{-1} where

$$\lambda(\gamma) = e^{t(\gamma)+i\phi(\gamma)}. \tag{2}$$

[*] drmlgentry@protonmail.com

The real part $t(\gamma) = \ell(\gamma)/2$ is half the complex translation length, and $\phi(\gamma)$ is the *twist angle* measuring the rotational component of the loxodromic motion.

B. A-factor data for m003

For the optimal PMNS manifold $m003$ (`OrientableClosedCensus`[1]) with word triple `aa`, `ab`, `aB`, the eigenvalue data is:

TABLE I. A-factor eigenvalue data for optimal m003 word triple. Translation lengths t and twist angles ϕ from $\lambda = e^{t+i\phi}$.

Word	t	ϕ (deg)	$e^{\lvert t\rvert}$
`aa`	-0.8894	-168.6	2.4338
`ab`	-0.4992	$+83.7$	1.6473
`aB`	-0.4447	$+95.7$	1.5601

III. CP PHASE PREDICTION FROM TWIST ANGLES

A. The geometric CP phase

The CP-violating phase δ_{CP} in the standard PMNS parametrization is a rephasing-invariant quantity measuring the imaginary part of products of mixing matrix entries. In the Iwasawa framework, the natural candidate is a combination of twist angles around the geodesic triangle formed by the three holonomy word axes.

Proposition 1 (CP phase from twist angles). *For the optimal PMNS word triple (`aa`, `ab`, `aB`) on manifold $m003$, the combination*

$$\delta_{\mathrm{geom}} = \phi_{aa} - \phi_{ab} + \phi_{aB} \tag{3}$$

predicts the CP phase δ_{CP}.

Numerically:

$$\begin{aligned}\delta_{\mathrm{geom}} &= (-168.6) - (83.7) + (95.7) \\ &= -156.6 \equiv 203.5 \pmod{360}.\end{aligned} \tag{4}$$

The PDG 2024 central value is $\delta_{\mathrm{CP}} = 197$, giving a discrepancy of 6.5 or 3.3%. This prediction involves no free parameters: the twist angles are fixed by the holonomy of $m003$, and the combination in (**??**) is the unique linear combination of three twist angles consistent with the signed sum around the geodesic triangle (positive for the "squared" word aa, negative for the cross-words).

B. Physical interpretation

The combination $\phi_{aa} - \phi_{ab} + \phi_{aB}$ has a natural geometric interpretation. The word $aa = a^2$ satisfies $\phi(a^2) = 2\phi(a)$ by additivity of twist angles (Proposition **??**), so $\phi_{aa}/2 = \phi_a = -84.3$ is the fundamental twist of the generator a. The combination becomes:

$$\delta_{\mathrm{geom}} = 2\phi_a - \phi_{ab} + \phi_{aB}, \tag{5}$$

which measures the signed deficit of twist angles around the generator triangle (a, b, B) on $m003$.

IV. THE ROW-PHASE OBSTRUCTION

Proposition 2 (Row-phase invariance of Jarlskog). *Let L be a lower-triangular matrix with unit diagonal and $L' = DL$ where $D = \mathrm{diag}(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3})$ is a diagonal phase matrix. If $L = QR$ is the QR decomposition, then $L' = (DQ)R$, so $Q \to DQ$. The Jarlskog invariant*

$$J = \mathrm{Im}(Q_{11}Q_{22}Q_{12}^*Q_{21}^*) \tag{6}$$

satisfies $J(DQ) = J(Q)$ for all diagonal phase matrices D. Consequently, multiplying L-entries by row phases cannot generate $J \neq 0$ from $J = 0$.

Proof. Under $Q \to DQ$: $Q_{ij} \to e^{i\alpha_i}Q_{ij}$. Then $Q_{11}Q_{22}Q_{12}^*Q_{21}^* \to e^{i\alpha_1}e^{i\alpha_2}e^{-i\alpha_1}e^{-i\alpha_2}Q_{11}Q_{22}Q_{12}^*Q_{21}^* = Q_{11}Q_{22}Q_{12}^*Q_{21}^*$. □

This theorem explains why naive phase multiplication of the real optimal L-entries produces $J = 0$ for all phase fractions. The CP phase must enter through *genuine off-diagonal* complex structure in L — not reducible to row phases.

V. COMPLEX BOREL CONSTRUCTION

A. Complex axis extraction

The real Borel construction uses only the real Pauli components of $\log\rho(\gamma)$:

$$\hat{n}^{(\mathrm{Re})}(w) = \frac{1}{\|\mathbf{n}\|}\mathbf{n}, \quad \mathbf{n} = (\mathrm{Re}[L_{01}+L_{10}], \mathrm{Im}[L_{10}-L_{01}], \mathrm{Re}[L_{00}-L_{11}]) \tag{7}$$

The *complex axis* incorporates the imaginary Pauli components:

$$\hat{n}^{(\mathrm{C})}(w;\alpha) = \frac{\hat{n}^{(\mathrm{Re})} + i\alpha\,\hat{n}^{(\mathrm{Im})}}{\|\hat{n}^{(\mathrm{Re})} + i\alpha\,\hat{n}^{(\mathrm{Im})}\|}, \tag{8}$$

where $\alpha \in [0, 1]$ interpolates between the real ($\alpha = 0$) and fully complex ($\alpha = 1$) constructions, and $\hat{n}^{(\mathrm{Im})}$ is the normalized imaginary part.

B. Complex L-entries and Jarlskog generation

With complex axes, the Borel L-entries become:

$$\ell_{ij} = \lambda \cdot (\hat{n}_i^{(\mathrm{C})*} \cdot \hat{n}_j^{(\mathrm{C})}), \quad i > j, \tag{9}$$

using the Hermitian inner product (conjugate of first argument). These ℓ_{ij} are genuinely complex — not reducible to row phases — and generate $J \neq 0$.

C. Numerical results

Table ?? summarizes the best results from the complex Borel construction scan on $m003$.

TABLE II. Best results from complex Borel construction on $m003$. α is the imaginary weight (??), $\mathcal{F}$ is permutation-minimized Frobenius fitness, J is the Jarlskog invariant. PDG targets: $\mathcal{F} = 0.01897$, $J = -0.00966$.

Words	α	scale	$\mathcal{F}$	J
aa/ab/aB (real)	0	1.0	0.01897	0.000
Ab/aB/aa	0.897	1.729	0.0489	−0.017
Ab/aa/aB	1.198	1.916	0.0931	+0.020

The best result (words `Ab/aB/aa`, $\alpha = 0.897$) achieves $J = -0.017$, which has the correct sign and is within a factor of 1.75 of the target $J_{\mathrm{PDG}} = -0.00966$. The fitness degradation from 0.019 to 0.049 reflects a genuine tension: the real construction optimizes $|U_{\mathrm{PMNS}}|$ at $\alpha = 0$ while the complex construction optimizes $J \neq 0$ at $\alpha \neq 0$.

VI. TRANSLATION LENGTH THEOREM

Proposition 3 (2:1 translation length ratio)**.** *For any element $\gamma \in \pi_1(M)$ and integer $n \geq 1$, the translation length satisfies $\ell(\gamma^n) = n\ell(\gamma)$. In particular, $\ell(aa) = 2\ell(a)$.*

Proof. The eigenvalue of $\rho(\gamma^n)$ is λ^n where $\lambda = e^{t+i\phi}$ is the eigenvalue of $\rho(\gamma)$. The translation length $\ell(\gamma) = 2|t|$, so $\ell(\gamma^n) = 2|nt| = n\ell(\gamma)$. □

Numerically for $m003$: $|t_{aa}|/|t_{aB}| = 0.88944/0.44472 = 2.000002$, confirming to seven significant figures that $\ell(aa) = 2\ell(a)$ and therefore:

[Isospectrality of a and aB] On manifold $m003$, the words a and aB have equal translation length: $\ell(a) = \ell(aB)$. This is a non-trivial constraint on the length spectrum of $m003$, reflecting the arithmetic structure of its fundamental group.

VII. PREDICTION: LENGTH-3 WORDS

The complex Borel scan over all length-2 word triples on $m003$ achieves a minimum joint loss $\mathcal{F} + c|J - J_{\mathrm{PDG}}|$ at fitness 0.049 and $J = -0.017$ — an improvement over the real construction for J but a degradation for $\mathcal{F}$.

Proposition 4 (Length-3 prediction)**.** *Simultaneous achievement of $\mathcal{F} \leq 0.020$ and $|J - J_{\mathrm{PDG}}| \leq 0.002$ in the complex Borel construction on $m003$ requires at least one word of length ≥ 3 in the optimal triple.*

This is supported by the exhaustive length-2 scan and constitutes a concrete testable prediction: a scan over length-3 words on $m003$ with complex axes should find a triple achieving both targets simultaneously.

VIII. DISCUSSION

A. Completion of the Iwasawa assignment

The three papers in this series now assign all three Iwasawa factors:

Factor	Physical content	Manifold
K	CKM quark mixing	$m006$, $H_1 = \mathbb{Z}/5$
N	PMNS lepton mixing	$m003$, $H_1 = \mathbb{Z}/5$
A	CP-violating phase	$m003$ (twist angles)

Both the CKM and PMNS manifolds have $H_1 = \mathbb{Z}/5$, and the CP phase arises from the A-factor of the *same* manifold as PMNS. This suggests that $m003$ encodes the entire lepton sector of the Standard Model, while $m006$ encodes the quark sector.

B. Quark-lepton asymmetry of CP violation from A-factor degeneracy

The A-factor twist angles of the optimal CKM words on $m006$ reveal a striking asymmetry with the PMNS case. For the optimal CKM word triple `aaB`, `AbA`, `AAb` on $m006$ (`OrientableClosedCensus`[43]), the A-factor eigenvalue data is:

TABLE III. A-factor eigenvalue data for optimal CKM words on $m006$. All three words are *isospectral*: they share the same translation length t and twist angle ϕ.

Word	t	ϕ (deg)	$e^{\vert t\vert}$
`aaB`	+0.8859	92.49	2.425
`AbA`	+0.8859	92.49	2.425
`AAb`	+0.8859	92.49	2.425

Proposition 5 (Triple isospectrality of CKM words on $m006$)**.** *The three optimal CKM words `aaB`, `AbA`, `AAb` all have the same holonomy eigenvalue $\lambda = e^{t+i\phi}$ with $t = 0.8859$ and $\phi = 92.49 \approx \pi/2$. They therefore represent geodesics of equal length with equal twist angle, lying in the same conjugacy class of* $\mathrm{PSL}(2,\mathbb{C})$.

This triple isospectrality is in sharp contrast to the PMNS case, where the three words `aa`, `ab`, `aB` have three distinct eigenvalues (see Table ??). The geometric consequence is immediate: since all three CKM twist angles are equal, every linear combination $c_1\phi_{\mathtt{aaB}} + c_2\phi_{\mathtt{AbA}} +$

$c_3\phi_{\texttt{AAb}}$ collapses to $(c_1+c_2+c_3)\phi_{\text{common}}$, giving no variation in the CP phase across different sign combinations.

[Geometric suppression of quark CP violation] The degenerate A-factor of $m006$ produces a CP phase that is independent of the sign pattern of the word combination. This degeneracy is geometrically consistent with the observed suppression $J_{\text{CKM}} \approx 3\times 10^{-5} \ll J_{\text{PMNS}} \approx 10^{-2}$: the quark manifold $m006$ has a nearly degenerate A-factor, while the lepton manifold $m003$ has a non-degenerate A-factor with three distinct twist angles.

The common twist angle $\phi \approx 92.49 \approx \pi/2$ is notable: $e^{i\phi} \approx i$, so the holonomy eigenvalue is nearly purely imaginary. We observe that $7\theta_{12}^{\text{CKM}} = 7\times 13.04 = 91.28$, within 1.2 of ϕ, suggesting a possible connection between the common CKM twist angle and the Cabibbo angle. Whether this is a numerical coincidence or reflects a deeper geometric structure of $m006$ is an open question.

C. Mass ratios and the A-factor spectrum

The translation lengths $e^{|t|}$ of the three optimal PMNS words are 2.434, 1.647, 1.560. These are not in the ratio of lepton masses ($m_\tau/m_\mu \approx 16.8$), indicating that mass ratios require either higher-length words or a different assignment of the A-factor spectrum. This is an open problem.

IX. CONCLUSION

We have identified the A-factor of the Iwasawa decomposition as the geometric source of CP violation in the hyperbolic flavor geometry framework. The key results are:

1. The twist angles of the optimal $m003$ word triple predict $\delta_{\text{CP}} = 203.5$, within 3.3% of the PDG value of 197, with no free parameters.

2. The real Borel construction has $J = 0$ identically by the row-phase invariance theorem, establishing a clean theoretical separation between the real ($|U|$-fitting) and complex (CP-generating) aspects of the construction.

3. The complex Borel construction generates $J \neq 0$ and achieves $J = -0.017$ at fitness 0.049, with the correct sign and within a factor of 1.75 of the PDG Jarlskog value.

4. The 2:1 translation length ratio $\ell(\texttt{aa}) = 2\ell(\texttt{aB})$ is proved as a theorem from the group law, with the non-trivial corollary that words a and `aB` are isospectral on $m003$.

5. Simultaneous achievement of $\mathcal{F} = 0.019$ and $J = J_{\text{PDG}}$ requires length-3 holonomy words — a concrete, falsifiable prediction for the next scan.

6. The three optimal CKM words `aaB`, `AbA`, `AAb` on $m006$ are *all* isospectral with common twist angle $\phi \approx \pi/2$. This triple isospectrality geometrically explains the suppression $J_{\text{CKM}} \ll J_{\text{PMNS}}$: the degenerate A-factor of $m006$ cannot generate the rich CP structure that the non-degenerate A-factor of $m003$ produces.

Together with [? ?], these results complete the assignment of all three Iwasawa factors to physical observables:

Factor	Observable	Source
K	CKM quark mixing	$m006$ real axes
N	PMNS lepton mixing	$m003$ real axes
A	CP phases + quark/lepton asymmetry	twist angles of $m003$

The quark-lepton asymmetry of CP violation — one of the outstanding puzzles of flavor physics — emerges as a consequence of the distinct isospectrality structures of $m003$ and $m006$: three distinct twist angles versus three degenerate twist angles.

ACKNOWLEDGMENTS

The author thanks the developers of SnapPy [?] for providing the hyperbolic manifold census and holonomy computation tools that made this numerical study possible.

Appendix A: Complex axis data for m003

Table ?? gives the full complex axis data extracted from $\log\rho(\gamma)$ for the six fundamental words on $m003$.

TABLE IV. Complex axis data for $m003$ words. $\hat{n}^{(\text{Re})}$ and $\hat{n}^{(\text{Im})}$ are unit vectors from real and imaginary Pauli decomposition of $\log\rho(\gamma)$.

Word	$\lvert\mathbf{n}_{\text{Re}}\rvert$	$\lvert\mathbf{n}_{\text{Im}}\rvert$
`aa`	15.252	16.250
`ab`	4.326	5.123
`aB`	13.291	13.675
`a`	8.567	9.153
`b`	3.948	3.934
`B`	3.948	3.934

Appendix B: A-factor data for m006 (CKM manifold)

Table ?? gives the complex axis data for the optimal CKM words on $m006$ (`OrientableClosedCensus`[43]). The near-equal imaginary norms confirm the near-degenerate A-factor structure.

TABLE V. Complex axis data for $m006$ optimal CKM words. All three words are isospectral with $\phi \approx 92.5$.

Word	$\lvert\mathbf{n}_{\mathrm{Re}}\rvert$	$\lvert\mathbf{n}_{\mathrm{Im}}\rvert$	$\lvert\mathbf{n}_{\mathrm{Im}}\rvert/\lvert\mathbf{n}_{\mathrm{Re}}\rvert$
`aaB`	3.518	4.434	1.260
`AbA`	21.782	21.949	1.008
`AAb`	12.934	13.213	1.022

For comparison, the PMNS manifold $m003$ has $|\mathbf{n}_{\mathrm{Im}}|/|\mathbf{n}_{\mathrm{Re}}|$ ratios of 1.065, 1.185, 1.029 for words `aa`, `ab`, `aB` respectively (Table **??**). The near-unity ratios for $m006$ reflect the $\phi \approx \pi/2$ twist angle: at exactly $\phi = \pi/2$ the real and imaginary Pauli norms would be equal.